

WENBO WANG

Email: wenbowang@g.harvard.edu; wenbowang1997@gmail.com

LinkedIn: <https://www.linkedin.com/in/wenbo-wang-9042a213a/>

Education

Harvard University, John A. Paulson School of Engineering and Applied Sciences Ph.D. student in Engineering Sciences Master of Science in Engineering Sciences <i>en route</i>	Cambridge, the United States Sept. 2019-Aug. 2026 (Expected) Jun. 2023
Zhejiang University, Department of Chemistry, Chu Kochen Honors College Bachelor of Chemistry, Qiushi Honors Program	Hangzhou, China Sep.2015-Jul.2019

Research Experience

Broad Institute of Harvard and MIT Research Assistant with Professor Xiao Wang	Cambridge, the United States Sep.2021-present
Harvard University, John A. Paulson School of Engineering and Applied Sciences Research Assistant with Professor Jia Liu	Boston, the United States Sep.2021-present
Westlake University, School of Life Sciences Visiting Student in School of Life Science with Professor Yi Sun	Hangzhou, China Sep.2019-June.2021
University of Utah, Department of Chemistry Research Scholar in Department of Chemistry with Professor Peter J. Stang	Salt Lake City, the United States Feb.2018-Sep.2018
Zhejiang University, Department of Chemistry Research Assistant in State Key Laboratory of Chemical Engineering, Department of Chemistry, Center for Chemistry of High-Performance & Novel Materials with Professor Feihe Huang	Hangzhou, China Oct.2016-Jul.2019

Research Interests

- Agentic AI systems for biological discovery
- AI-driven modeling and interpretation of cellular organization and function
- *In situ* multi-omics for understanding biological systems across molecular, cellular, and tissue scales

Selected Publications

- **Wang, W. #**, Swain, S. #, Lee, J. #, Lin, Z. #, ... & Liu, J.* (2025). Agentic Lab: An Agentic-physical AI system for cell and organoid experimentation and manufacturing. Preprint at *bioRxiv*, in review at *Nature*.
- **Wang, W.#**, Lin, Z.#, ..., Alvarez-Dominguez, J.* & Liu, J.* (2025). Spatial transcriptomics AI agent charts hPSC-pancreas maturation in vivo. Preprint at *bioRxiv*, in review at *Nature Methods*.
- Li, Qiang#, Liu, R.#, Lin, Z.#, Zhang, X.#, **Wang, W.#**, ..., Alvarez-Dominguez, J.* & Liu, J.* (2025). Cyborg islets: implanted flexible electronics reveal principles of human islet electrical maturation. *Science*, 391(6787), eae3295.
- Lee, J.#, **Wang, W.#**, Li, Q., Lin, Z., Liu, R., Tang, Z., ... & Liu, J.* (2025). *In situ* graphene-seq: Spatial transcriptomics and chronic electrophysiological characterization of tissue microenvironments. *Nature Communications*, 2026.
- Lee, J.#, Lin, Z. #, **Wang, W.#**, ... Liu, J.* (2026). DeviceAgent: An autonomous multimodal AI agent for flexible bioelectronics. Preprint at *bioRxiv*, Preprint at *bioRxiv*, in review at *Nature Electronics*
- Lin, Z.#, **Wang, W.#**, Liu, R.#, Li, Q.#, Lee, J., Hirschler, C., & Liu, J.* (2025). Cyborg organoids integrated with stretchable nanoelectronics can be functionally mapped during development. *Nature Protocols*, 1-32.
- Sessler, C. D. #, Zhou, Y. #, **Wang, W. #**, Hartley, N. D., Fu, Z., Graykowski, D., ... Wang, X.* & Liu, J.* (2022). Optogenetic polymerization and assembly of electrically functional polymers for modulation of single-neuron excitability. *Science Advances*, 8(49), eade1136.
- **Wang, W.#**, Sessler, C. D.#, Wang, X., & Liu, J.* (2024). In Situ Synthesis and Assembly of Functional Materials and Devices in Living Systems. *Accounts of Chemical Research*, 57(15), 2013-2026.
- *Corresponding authors.
#Equal contribution.

For a full publication list, please see: <https://scholar.google.com/citations?user=TIyfmcgAAAAJ&hl=en>

Awards and Recognitions

- Zhejiang Provincial Scholarship, 2015-2016
- Scholarship for Outstanding Students, 2015-2017
- Excellent Student Scholarship in Zhejiang University, 2015-2017
- Scholarship for Outstanding Students in Basic Science, 2016-2018
- Research and Innovation Scholarship, 2017-2018
- Harvard Student Recognition of Teaching, 2022-2023
- Regeneron prize for creative innovation, Harvard nominee (two students each year), 2024
- Sky9 Fellowship, 2026

Invited Talks and Presentations

- The Harvard Materials Research Science and Engineering Center (MRSEC) seed meeting, Jan. 2023
- Human Islet Research Network (HIRN): STAgent: Autonomous spatial transcriptomics AI agent for Pancreatic Research and Beyond, July. 2025
- 21st Annual Broad Institute Scientific Retreat, July. 2025
- Guest Presenter, Smilow Center for Translational Research, University of Pennsylvania, Jan. 2025
- Guest Presenter, Broad Institute of MIT and Harvard, Dec. 2022

Professional Service

- Fellow, Agentic and Physical AI for Excellence in Science and Engineering, Harvard University, 2025-present
- Core team member, Biocosmo, 2024-present
- Member, American Chemical Society, 2025
- Poster judge, 21st Annual Broad Institute Scientific Retreat, Dec. 2025
- Harvard Topics in Bioengineering (TIB) Student Coordinator, 2023-2024
- Pre-Concentration Academic Advisors of Harvard University, 2024-2025
- Reviewer for journals and conferences: *PeerJ Computer Science*, *Heliyon*, *Gene Reports*, *Molecular Therapy*, *Journal of Cellular and Molecular Medicine*, *Biomedicine & Pharmacotherapy*, *BMC Bioinformatics*, *Frontiers in Immunology*, *Expert Systems*, *Open Medicine*, *Physiological Measurement*, *Biofabrication*

Teaching and Mentorship Experience

- BE131 Neuroengineering, 2022
 - Led weekly discussion sections, managed grading, and provided instruction to undergraduates.
 - Hosted weekly lab sessions and designed lab protocols for tissue clearance and fluorescence microscopy imaging.
 - Harvard Recognition of Teaching Award for outstanding teaching.
- Graduate student, Jessica Liu (Bioengineering), Harvard University, 2025-2026
- Fellow, Simran Swain (Bioengineering), Harvard University, 2025-2026
- Fellow, Bradley Altman (Chemistry), Harvard University, 2026-present
- Visiting undergraduate, Kaile Zhang (Applied Mathematics), Boston University, 2026-present
- Visiting master's student, Chuan Qiu (Computer Science), Northeastern University, 2026-present
- Master's student, Bradley Canales (Bioengineering), Harvard University, 2025-present
- Visiting undergraduate student, Chystal (Jingjing) Duan (Chemistry), Caltech, 2026-present
- Visiting undergraduate student, Shulei Peng (Chemistry), Fudan University, 2026-present

References

Professor Jia Liu (Thesis co-advisor)

Assistant Professor of Bioengineering
School of Engineering and Applied Sciences, Harvard University
Email: jia_liu@seas.harvard.edu

Professor Xiao Wang (Thesis co-advisor)

Thomas D. and Virginia Cabot Associate Professor of Chemistry; Core Member at Broad Institute
Department of Chemistry, Massachusetts Institute of Technology
Email: xwangx@mit.edu; wangxiao@broadinstitute.org

Professor Na (Lina) Li (collaborator)

Winokur Family Professor of Electrical Engineering and Applied Mathematics
School of Engineering and Applied Sciences, Harvard University
Email: nali@seas.harvard.edu

Professor Juan R. Alvarez-Dominguez (collaborator)

Assistant Professor of Cell and Developmental Biology
Perelman School of Medicine, University of Pennsylvania
Email: juan.alvarez@pennmedicine.upenn.edu